Homework 12: 10/20 Content

MATH 1271, Garrett Credi

10/20/2026

Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Derivative of Inverse Functions

Choose 5 of the following problems to solve.

1. AC 2.6.1-10 (each of these counts as one problem)
2. At most one of APEX 2.7.5-8
3. APEX 2.7.9-30 (each of these counts as one problem)

2 Type 2 - Implicit Differentiation

Choose 5 of the following problems to solve.

1. AC 2.7.1-13 (each of these counts as one problem)

2. APEX 2.6.13-39 (each of these counts as one problem)

3. PAULS-ASSIGN.Derivatives.Implicit Differentiation.1-31 (each of these counts as one problem)

3 Challenge Problem

Assuming that $f(x)$ is a differentiable function, at which points is $f^{-1}(x)$ differentiable?

Assuming it's second differentiable, at which points is $f^{-1}(x)$ second differentiable?

What about third differentiability?

Use the above work to guess a condition that would guarantee the existence of *all* derivatives of $f^{-1}(x)$, assuming all derivatives of $f(x)$ exist.